Master Theoretical & Computational Handout

Data-Driven Modeling of Galactic Rotation Curves with Artificial Neural Networks

arXiv:2404.05833v2 Full Reproduction & Independent Validation

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Abstract

This comprehensive master handout details the physical theory, mathematical intuition, deep learning architecture, and computational implementation for reproducing arXiv:2404.05833v2. It covers galactic kinematic decomposition, Monte Carlo Dropout uncertainty estimation, Genetic Algorithm hyperparameter search, and physical classification of 17 THINGS galaxies into Groups A, B, and C.

1 Astrophysical Foundations

1.1 Rotational Equilibrium

In spiral galaxies, circular orbits in the equatorial plane obey:

$$\frac{V^2(r)}{r} = -\frac{\partial \Phi(r)}{\partial r} = \frac{GM(<r)}{r^2} \tag{1}$$

1.2 Mass Components

The total observed velocity is decomposed as:

$$V^2_{\text{total}}(r) = V^2_{\text{gas}}(r) + \Upsilon_\star V^2_\star(r) + V^2_{\text{halo}}(r) \tag{2}$$

where V_{gas} is from 21-cm HI data ($1.4\times$ helium/metal correction), $V_\star$ is from $3.6\,\mu$m Spitzer surface brightness, and V_{halo} is modeled via the Navarro-Frenk-White (NFW) profile:

$$\rho_{\text{NFW}}(r) = \frac{\rho_i}{\left(\frac{r}{R_s}\right)\left(1+\frac{r}{R_s}\right)^2} \tag{3}$$

2 Machine Learning & Mathematical Formulation

2.1 Multi-Output Neural Network

An ANN maps radial distance $r \to [V(r), \sigma_V(r)]$ without physical priors.

2.2 Monte Carlo Dropout (MC-DO)

Dropout ($p = 0.3$) is retained active during inference across $N_{\text{MC}} = 100$ forward passes:

$$\bar{V}(r) = \frac{1}{N_{\text{MC}}} \sum_{t=1}^{N_{\text{MC}}} \hat{V}^{(t)}(r) \tag{4}$$

$$\sigma^2_{\text{MC}}(r) = \frac{1}{N_{\text{MC}}} \sum_{t=1}^{N_{\text{MC}}} \left(\hat{V}^{(t)}(r) - \bar{V}(r)\right)^2 \tag{5}$$

$$\sigma_{\text{total}}(r) = \sqrt{\sigma^2_{\text{MC}}(r) + \bar{\sigma}^2_{\text{stat}}(r)} \tag{6}$$

3 Group Classification & Results

Group	Galaxies	Criterion
A	DDO154, NGC2903, NGC3621, NGC2403	NFW fits b
B	NGC3031, NGC2841, NGC3198, NGC5055	ANN betwe
C	NGC4736, NGC925, NGC6946, NGC7793, NGC2976, NGC7331, NGC3521, NGC2366, IC2574	ANN best

Table 1: Classification of 17 THINGS galaxies from arXiv:2404.05833v2.